Alireza Karimi

Gmail University Email Number LinkedIn Github Webpage

Education

University of Tehran, Bachelor's degree, Computer Software Engineering — Aug 2022 – Present

- Score: 18.17 (=3.8/4 GPA)

National Organization for Development of Exceptional Talents (Sampad), High and Middle School Diploma, Mathematics and Physics — Sep 2016 – Jun 2022

Publication and Research Experience

Research Assistant – Distributed Machine Learning Systems — Sep 2025 – Oct 2026

Supervisors: Prof. A. Khonsari and Dr. M. Dolati
This research resulted in the following manuscript:

"Compute and IO-Aware Distributed Training Management in Converged Networks"
Seyyed Ali Aghamiri, **Alireza Karimi**, Ahmad Khonsari, and Mahdi Dolati.
Submitted to IEEE Transactions on Cloud Computing (TCC).

Formulated the joint scheduling problem for multi-job distributed deep learning training over shared networks as a time-indexed Mixed-Integer Linear Programming (MILP) model, accounting for computation, GPU memory, storage I/O, and network contention.

Implemented and evaluated the framework using the FlexNet and FlexNetPacket simulators from TopoOpt

Research Interests

Distributed Machine Learning Systems
Distributed Deep Learning
Distributed or Decentralized Systems
Distributed and Network Optimization
Federated Learning

Selected Projects

Distributed ML Systems – Implemented distributed ML workloads using MPI and federated learning, CUDA and PyTorch DDP, Spark/HDFS, and communication-efficient distributed SGD.

Distributed Systems – Worked on IPC and concurrency, multi-VM gRPC services, and a replicated key-value store supporting strong and eventual consistency with failure and network-delay simulation.

Computer Networks – Worked on a TCP-based HTTP server, reliable data transfer, distance-vector routing, ICMP tools, a P4 learning switch, and TCP congestion-control simulations.

Data Science – Developed data-intensive ML pipelines using Kafka and Spark, semantic search, deep learning, LoRA-based fine-tuning, and large-scale demand forecasting.

Artificial Intelligence – Implemented transfer learning and GANs, clustering and decision trees, reinforcement learning, and RNN/LSTM models across multiple datasets.

Technical Skills

Distributed ML Systems: PyTorch DDP, CUDA, Apache Spark, Apache Kafka, HDFS, SLURM, Docker

Networking & Optimization: Wireshark, ILP, SCIP

Programming Languages: C++, C, Python, Go, Java, SQL, MATLAB, Verilog

Development Tools: Git, GitHub, Jupyter Notebook, Makefile

Currently Learning: Ray, Kubernetes, P4

TA Experience

Discrete Mathematics, Sep 2023 – Feb 2025
Teaching Assistant, under the supervision of Prof. Mohammadi

Introduction to Computing Systems and Programming (Lab), Sep 2023 – Feb 2026
Teaching Assistant, under the supervision of Prof. Hashemi and Prof. Moradi

Introduction to Computing Systems and Programming, Feb 2024 – Jul 2024
Teaching Assistant, under the supervision of Prof. Dousti

Selected Coursework

Distributed Machine Learning Systems – 17.5/20
Computer Networks – 20/20
Algorithm Design – 20/20
Foundations of Data Science – 19.3/20
Database Design – 18.8/20
Discrete Mathematics – 19.25/20
Advanced Programming – 20/20
Introduction to Distributed Computing – 20/20
Computer Networks Laboratory – 20/20
Data Structures and Algorithms – 20/20
Artificial Intelligence – 18.5/20
Software Engineering – 19.55/20
Engineering Probability and Statistics – 18.6/20

Honors

Iranian National University Entrance Examination
169th / 140,000+ – Mathematics and Physics
403rd / 190,000+ – English

References

Prof. Ahmad Khonsari | University of Tehran, Iran | Email | Webpage
Dr. Mahdi Dolati | Sharif University of Technology, Iran | Email | Webpage
Dr. Siamak Mohammadi | University of Tehran, Iran | Email | Webpage